DeskSense

An on-device AI desk wellness companion that never uploads your face

TECHNOVA 2026 — National AI Innovation Challenge · Innovation Summary · Team «TEAM NAME», «College» · Domain: Healthcare / Occupational Wellbeing · SDG 3 (primary), SDG 8, SDG 9 · Stage: Working Prototype

The problem

Desk work is the default for millions of Indian students and knowledge workers, and it harms them in two slow, invisible ways. **Physically**, posture degrades gradually across a working day; the person notices only at night, as neck and lower-back pain. Work-related musculoskeletal pain is widely documented among Indian desk workers, and it begins in college, where students already sit six to nine hours a day. **Attentionally**, focus leaks away in thirty-second slices — a phone check, a tab switch — that feel insignificant alone and are impossible to recall by evening. Without measurement, nothing changes.

Tools addressing this exist, but they are built wrong for the person experiencing the problem. Almost all stream webcam video to a remote server, and most are sold to employers rather than to the individual being watched. They also measure the wrong things — keystrokes, application usage, idle time — none of which describe the body. The person who most needs this information must either accept surveillance to obtain it, or go without.

The solution

DeskSense is a self-contained desk device built on the Arduino UNO Q. One USB camera watches the workstation and every computation happens on the board. Once per second the system determines four things — whether a person is present, whether a phone is in frame, whether the person is facing their screen, and how far their posture has drifted from their own calibrated baseline — and combines them into an engagement score, a posture score, and a state (engaged / distracted / away).

The output is deliberately physical rather than another notification. A 16×2 LCD on the desk shows live scores. Relay channels switch a lamp on presence and fire a posture alert after ten continuous seconds below threshold, so feedback arrives in the room rather than on the screen the user is already losing to. A dashboard served by the board on the user's own network shows the session timeline, and an on-device language model writes a plain-English end-of-day summary. Posture calibration is automatic and personal: eight seconds of observation when the user sits down, median-locked, re-captured every time they return — so the score reflects that person in that chair, not a population average.

How artificial intelligence is used

Component	Model / method	Design decision behind it
Presence & phone detection	YOLOX-nano (COCO), via App Lab's Object Detection Brick, on the QRB2210 with Adreno GPU acceleration	Asymmetric thresholds — person 0.55, phone 0.15 — because a partly occluded phone flickers below confidence in a way a seated person does not
Gaze	ResNet-10 SSD face detector via OpenCV's DNN module; frontal face within 32% of frame centre	Replaced MediaPipe Pose, which crashes on this board: its prebuilt ARM wheels assume CPU extensions the Cortex-A53 does not implement
Posture	Bounding-box drift from an 8-second median baseline — head drop, box shrink, lateral shift, each weighted	Comparative rather than absolute. We claim drift detection, not joint-angle estimation, because that is what one box can support
Decision layer	Debouncers (symmetric for gaze, asymmetric hold for phone) plus exponential moving averages on both scores	A confirmed look-away and an undetected face are penalised differently — the second is ambiguous, not evidence
Summarisation	Qwen 3.5 0.8B, 4-bit quantised, running locally via llama.cpp	Converts aggregate rows into readable feedback with no cloud call and no account

What makes it different

It belongs to the person it watches. Commercial focus and posture tools are

It acts on the physical world. Lamp, LCD and alert indicator — enabled by the UNO

It is honest about its limits. Posture is a calibrated proxy. Gaze is coarse and

sold to employers and stream video to vendor servers. DeskSense is a device the user buys and controls. Privacy here is an architectural property, not a policy promise.

Q's dual-processor design: heavy vision on the Linux side, hard real-time I/O on the microcontroller, linked by RPC.

degrades under backlight. A system overclaiming medical accuracy would be easy to build and impossible to defend.

Privacy — the precise boundary

True: no frame is ever written to disk; no frame, score or summary reaches any cloud service or third party; detection, gaze, posture and the daily summary all run on the board; only 5-minute aggregate rows persist, in a local SQLite file; the language model is local, quantised and offline.

Stated openly: the dashboard pushes a live JPEG over WebSocket to browsers on the same LAN — that is how the preview works. Those frames are never stored, but the dashboard has no authentication yet. Adding a shared-token check is the next commit and closes the gap.

Beneficiaries and impact

Users: students and knowledge workers at fixed desks for six or more hours a day; college labs and libraries, where one shared unit gives feedback without recording or identifying anyone; and small employers with a genuine duty of care around ergonomics but no appetite for surveillance software. Explicitly *not* built for employers who want to monitor staff — the architecture makes that impossible, since no data leaves the device the worker controls.

Economic	Social	Environmental
Musculoskeletal pain drives absenteeism and treatment cost; early feedback is far cheaper than later physiotherapy. Local inference means no cloud bill and no per-user server cost, so unit economics do not degrade as users are added. Hardware is under ₹12,000 at prototype scale against recurring per-seat subscriptions.	Self-knowledge about your own health should not cost you your privacy; today the two are bundled. It also reaches people with no formal ergonomic assessment at all — students in hostel rooms, workers at kitchen tables — who are precisely the highest-risk group.	Inference runs on one low-power board rather than a data centre, so per-user marginal energy and bandwidth for the AI is effectively zero. Components are standard and separable — a failed relay or LCD is replaced, not the whole unit. The lamp relay also releases on absence, cutting routine wasted lighting.

Technical profile

Hardware: Arduino UNO Q (Qualcomm QRB2210 quad-core Cortex-A53 + Adreno GPU, Debian Linux; STM32U585 Cortex-M33) · USB webcam · 3-channel opto-isolated relay module, active-low, separate 5 V coil supply · 16×2 I²C LCD (PCF8574 / hd44780)
Software: Arduino App Lab (WebUI, Object Detection, LLM Bricks) · Python · OpenCV · YOLOX-nano · llama.cpp · SQLite · FastAPI · Router Bridge RPC
Measured behaviour: 1 s scoring cadence · ~6.7 fps live preview · 12–19 s slouch-to-relay latency · 96 stored rows per 8-hour day · engagement EMA reaches 90% in 4.5 s, posture in 8.0 s
Datasets: none collected. Pretrained COCO and face-detection models; posture calibrates per user at runtime.
Stage: working prototype, running daily · **Patent:** not filed · **Startup status:** student project

Roadmap

Now — token auth on the dashboard, closing the one gap we disclose. **3 months** — pilot in a college lab with roughly 20 users, measuring adoption and retention rather than accuracy alone. **6 months** — enclosure, consolidated BOM, sub-ten-minute install. **12 months** — weekly summaries, and a real pose model if one becomes available for ARMv8.0.

We are seeking mentoring, product-development support, industry connections for a pilot cohort, and incubation runway to test whether people will actually buy this.

Source code: github.com/omprakashrajp75-ui/desksense · Team «TEAM NAME» · «College» · Contact: «email» · Previously entered in the Arduino Physical AI Challenge India 2026.